

"Considerate la vostra semenza:
fatti non foste a viver come bruti,
ma per seguir virtute e canoscenza"
(Dante Alighieri, Inferno XXVI)

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1 Introduction

In 1827, the Scottish botanist Robert Brown was investigating the fertilization process in *Clarkia pulchella*, then a newly discovered species of flowering plant, when he observed under a microscope a “rapid oscillatory motion” of microscopic particles within pollen grains suspended in water. Initially, he believed that this motion was a vital activity peculiar to the male reproductive cells of plants. However, he later observed the same movement in pollen from plants that had been dead for over a century. Further studies revealed that this motion occurred not only in particles of other organic substances but also in fragments of glass and particles of smoke. He thus demonstrated that the phenomenon was not exclusive to living matter.

Early explanations attributed the motion to thermal convection currents in the fluid, but this hypothesis was abandoned because nearby particles exhibited entirely uncorrelated movements. By the 1860s, theoretical physicists had become increasingly interested in Brownian motion (BM) and were seeking a consistent explanation for its characteristics: a particle appeared equally likely to move in any direction, its subsequent motion seemed unrelated to its previous motion, and the movement never ceased. A study in 1889 showed that smaller particles and lower viscosity of the surrounding fluid resulted in faster motion. In 1887, it had been suggested that the direct relationship between temperature and the speed of Brownian motion arose from “thermal molecular motion in the liquid environment”.

The kinetic theory of gases, developed by physicists such as James Clerk Maxwell, Rudolf Clausius, and Ludwig Boltzmann, describes molecules as being in constant motion, colliding with one another and moving back and forth. The temperature of a substance is therefore related to the average kinetic energy of its molecules. It was natural to suppose that this motion could be transferred to larger particles visible under a microscope. In 1905, Albert Einstein developed a quantitative theory of Brownian motion. This theory led to an early and compelling confirmation of the molecular theory of matter through Jean Perrin’s experiments to determine Avogadro’s number, for which Perrin was awarded the Nobel Prize in 1926.

In 1923, Norbert Wiener established the mathematical existence and construction of Brownian motion. The terms Brownian motion and Wiener process refer to the

same process, although the former is commonly used when discussing the physical phenomenon and the latter when referring to the mathematical model.

A well-known application of the theory of the Wiener process is the Black–Scholes model, formulated by Fischer Black and Myron Scholes in their 1973 article, “The Pricing of Options and Corporate Liabilities”. This model was the first major model for pricing financial derivatives. The Black–Scholes equation is a parabolic partial differential equation (PDE) that describes the evolution of a derivative’s price function in terms of its partial derivatives with respect to time and the underlying asset price. The Black–Scholes formulas provide the prices of European call and put options. They can be obtained by solving the PDE subject to the terminal condition that the derivative’s price at maturity equals its payoff.

In this model, the stock price process S evolves under the risk-neutral probability measure $\mathcal{Q}$ according to the following stochastic differential equation:

$$dS(t) = rS(t) dt + \sigma S(t) dW^*(t), \tag{1.1}$$

where W^* is a standard Wiener process under $\mathcal{Q}$. The process S is a geometric Brownian motion (GBM).

2 Mathematical Background

This chapter introduces the probabilistic concepts and stochastic calculus tools needed to develop the Black–Scholes model. The presentation follows primarily [Björk, 2009] and [Oksendal, 2013].

2.1 Brownian Motion

We begin by describing the probability space on which our stochastic processes are defined and the information available as time evolves.

Definition 1. A σ -algebra on a nonempty set Ω is a collection $\mathcal{F}$ of subsets of Ω satisfying the following properties:

1. $\Omega \in \mathcal{F}$;
2. if $A \in \mathcal{F}$, then $A^c = \Omega \setminus A \in \mathcal{F}$;
3. if $A_n \in \mathcal{F}$ for every $n \in \mathbb{N}$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

The pair $(\Omega, \mathcal{F})$ is called a measurable space.

These properties also imply that $\emptyset \in \mathcal{F}$. The elements of $\mathcal{F}$ are called events. Given two σ -algebras $\mathcal{F}$ and $\mathcal{F}'$ on Ω , we say that $\mathcal{F}'$ is finer than $\mathcal{F}$ if $\mathcal{F} \subseteq \mathcal{F}'$. In probabilistic terms, a finer σ -algebra represents at least as much information.

Definition 2. A probability measure on $(\Omega, \mathcal{F})$ is a function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ such that $\mathbb{P}(\Omega) = 1$ and, for every sequence of pairwise disjoint events $(A_n)_{n \geq 1}$,

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

In particular, $\mathbb{P}(\emptyset) = 0$. Unless stated otherwise, all expectations and almost-sure statements below refer to $\mathbb{P}$. In the financial application, $\mathbb{P}$ represents the physical probability measure.

Definition 3. A filtration on $(\Omega, \mathcal{F}, \mathbb{P})$ is a family $\mathcal{M} = (\mathcal{M}_t)_{t \geq 0}$ of sub- σ -algebras of $\mathcal{F}$ such that

$$\mathcal{M}_s \subseteq \mathcal{M}_t \quad \text{whenever } 0 \leq s \leq t.$$

The σ -algebra $\mathcal{M}_t$ represents the information available at time t . The inclusion above expresses the fact that information is not lost over time. On a finite horizon $[0, T]$, $\mathcal{M}_T$ contains the information available by time T ; it need not contain every subset of Ω . The choice $\mathcal{M}_0 = \{\emptyset, \Omega\}$ describes a model with no nontrivial initial information, but it is not part of the definition of a filtration.

For a stochastic process X , its natural filtration is

$$\mathcal{M}_t^X = \sigma(X(s) : 0 \leq s \leq t).$$

It records the information generated by observing X up to time t . A random variable Z is $\mathcal{M}_t^X$ -measurable if, for every Borel set $A \subseteq \mathbb{R}$, the event $\{Z \in A\}$ belongs to $\mathcal{M}_t^X$. Informally, its value can be determined from the information available at that time. We write that Z is measurable with respect to the σ -algebra, rather than writing $Z \in \mathcal{M}_t^X$: a σ -algebra contains events, not random variables.

Definition 4. A real-valued stochastic process $X = (X(t))_{t \geq 0}$ is adapted to $\mathcal{M}$ if $X(t)$ is $\mathcal{M}_t$ -measurable for every $t \geq 0$.

Adaptedness means that the process does not use future information. For stochastic calculus, we work on a complete probability space and take the filtration to satisfy the usual conditions: it is right-continuous and contains all $\mathbb{P}$ -null events of $\mathcal{F}$ at time zero. A natural filtration can be completed and made right-continuous to obtain its usual augmentation. When working with the Brownian filtration below, we use this augmentation, still denoted by $\mathcal{M}^W$. We now define the (standard) Brownian Motion or (standard) Wiener process and some of its variants.

Definition 5. An adapted process $W = (W(t))_{t \geq 0}$ is a standard Brownian motion, or Wiener process, with respect to $\mathcal{M}$ under $\mathbb{P}$ if:

1. $W(0) = 0$ almost surely;
2. its sample paths $t \mapsto W(t, \omega)$ are continuous for almost every $\omega \in \Omega$;

3. for $0 \leq s < t$, the increment $W(t) - W(s)$ is independent of $\mathcal{M}_s$;
4. for $0 \leq s < t$,

$$W(t) - W(s) \sim \mathcal{N}(0, t - s).$$

The third property implies that increments over disjoint time intervals are independent. The fourth implies that their distribution depends only on the length of the interval. Thus Brownian motion has stationary, independent increments.

For $h > 0$,

$$W(t + h) - W(t) \sim \mathcal{N}(0, h). \quad (2.1)$$

Consequently,

$$\mathbb{E}[|W(t + h) - W(t)|^2] = h \longrightarrow 0 \quad \text{as } h \downarrow 0.$$

This establishes mean-square continuity and hence continuity in probability. It does not, by itself, establish continuity of sample paths. Path continuity is included in the definition above; when constructing Brownian motion, the fourth-moment identity

$$\mathbb{E}[|W(t) - W(s)|^4] = 3|t - s|^2$$

allows one to obtain a continuous modification using the Kolmogorov continuity theorem. Although Brownian paths are almost surely continuous, they are almost surely nowhere differentiable. To see why ordinary differentiation is problematic, note that, for each fixed t and $h > 0$,

$$\frac{W(t + h) - W(t)}{h} \sim \mathcal{N}\left(0, \frac{1}{h}\right).$$

The variance increases without bound as $h \downarrow 0$. This calculation illustrates the irregularity of the paths; every trajectory switches from increasing to decreasing trends with such speed that we would observe the same behavior if we could observe it more closely. This, in fact, makes Brownian Motion paths fractals.

Definition 6. A Brownian motion with drift, also called a generalized Brownian motion, is a process of the form

$$B(t) = \mu t + \sigma W(t), \quad t \geq 0, \quad (2.2)$$

where $\mu \in \mathbb{R}$, $\sigma > 0$, and W is a standard Brownian motion.

The process starts at zero, has stationary and independent increments, and satisfies

$$B(t) \sim \mathcal{N}(\mu t, \sigma^2 t).$$

Here μ is the drift rate and σ^2 is the variance rate. A nonzero initial value can be incorporated by adding a constant to B .

The exponential construction in [Ross, 2014] leads to geometric Brownian motion.

Definition 7. Let $B(t) = \alpha t + \sigma W(t)$, where $\alpha \in \mathbb{R}$ and $\sigma > 0$. For $x_0 > 0$, the process

$$X(t) = x_0 e^{B(t)} = x_0 e^{\alpha t + \sigma W(t)}, \quad t \geq 0, \quad (2.3)$$

is called a geometric Brownian motion.

In this parametrization, α is the drift of $\log X(t)$, not the drift coefficient of X in its stochastic differential equation. We will show that the latter is $\mu = \alpha + \sigma^2/2$. This distinction is important when modelling asset prices. Geometric Brownian motion remains strictly positive, and its value at each positive time has a lognormal distribution, in accordance with the non possible negativity of stock prices and normally distributed financial logarithmic returns.

2.2 Stochastic Integral

To describe the evolution of an asset price in continuous time, we need to combine a drift term with random fluctuations. For a process X governed by a diffusion model, a first-order approximation over a short interval is

$$\begin{aligned} X(t + \Delta t) - X(t) &\approx \mu(t, X(t))\Delta t \\ &\quad + \sigma(t, X(t))\Delta W(t), \end{aligned} \quad (2.4)$$

where

$$\Delta W(t) = W(t + \Delta t) - W(t) \sim \mathcal{N}(0, \Delta t). \quad (2.5)$$

The increment $\Delta W(t)$ is independent of $\mathcal{M}_t$. The deterministic functions μ and σ are the drift and diffusion coefficients. Their values at $(t, X(t))$ are known at time t , whereas

the Brownian increment is not. The two contributions in (2.4) therefore represent a drift determined by the current state and a random fluctuation whose magnitude is scaled by the diffusion coefficient.

The corresponding continuous-time model is the stochastic differential equation

$$\begin{cases} dX(t) = \mu(t, X(t)) dt + \sigma(t, X(t)) dW(t), \\ X(0) = x_0. \end{cases}$$

This notation is shorthand for the integral equation

$$X(t) = x_0 + \int_0^t \mu(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dW(s). \quad (2.6)$$

The first integral is an ordinary time integral. The second requires a stochastic integration theory. We could see the latter as a Riemann-Stieltjes integral for each Brownian path; however, the problem is that such paths have infinite total variation on every nontrivial interval, almost surely. Therefore, we need a different theory here.

We construct the Itô integral on a finite interval $[a, b]$ using square-integrable processes. Let $\mathcal{L}^2[a, b]$ denote the class of progressively measurable processes g , with respect to the Brownian filtration, such that

$$\mathbb{E} \left[\int_a^b |g(t)|^2 dt \right] < \infty,$$

meaning the expected total energy of the process is finite.

Progressive measurability means that, for every $u \in [a, b]$, the restriction $(t, \omega) \mapsto g(t, \omega)$ to $[a, u] \times \Omega$ is measurable with respect to $\mathcal{B}([a, u]) \otimes \mathcal{M}_u^W$. This condition ensures both adaptedness and the joint measurability needed for integration. In particular, every adapted process with continuous paths is progressively measurable.

As it is usually done in Measure Theory, we first consider an elementary adapted process of the form

$$g(t) = \sum_{k=0}^{n-1} \xi_k \mathbf{1}_{[t_k, t_{k+1})}(t), \quad a = t_0 < t_1 < \cdots < t_n = b,$$

where each ξ_k is square-integrable and $\mathcal{M}_{t_k}^W$ -measurable. Values at the finitely many

endpoints do not affect the integral. We define

$$\int_a^b g(t) dW(t) = \sum_{k=0}^{n-1} \xi_k (W(t_{k+1}) - W(t_k)). \quad (2.7)$$

The coefficient on each interval is known at its left endpoint. This non-anticipating choice is a defining feature of the Itô integral. For a general $g \in \mathcal{L}^2[a, b]$, the construction proceeds as follows:

1. Choose a sequence of elementary adapted processes g_n such that

$$\mathbb{E} \left[\int_a^b |g_n(t) - g(t)|^2 dt \right] \longrightarrow 0.$$

2. Define $Z_n = \int_a^b g_n(t) dW(t)$. The Itô isometry for elementary processes gives

$$\mathbb{E}[|Z_n - Z_m|^2] = \mathbb{E} \left[\int_a^b |g_n(t) - g_m(t)|^2 dt \right].$$

Hence (Z_n) is Cauchy in $L^2(\Omega, \mathcal{F}, \mathbb{P})$ and has a limit in that space.

3. Define the integral by this mean-square limit:

$$\int_a^b g(t) dW(t) = \lim_{n \rightarrow \infty} \int_a^b g_n(t) dW(t) \quad \text{in } L^2(\Omega). \quad (2.8)$$

The isometry also ensures that the limit does not depend on the approximating sequence.

Proposition 1. *Let $g \in \mathcal{L}^2[a, b]$ and define*

$$I(t) = \int_a^t g(u) dW(u), \quad a \leq t \leq b.$$

Then:

1. $\mathbb{E}[I(t)] = 0$;
2. *the Itô isometry holds:*

$$\mathbb{E}[I(t)^2] = \mathbb{E} \left[\int_a^t |g(u)|^2 du \right];$$

3. I has a continuous version adapted to $\mathcal{M}^W$;
4. I is a square-integrable martingale:

$$\mathbb{E}[I(t) \mid \mathcal{M}_s^W] = I(s), \quad a \leq s \leq t \leq b.$$

Since $I(t)$ has mean zero, its second moment in the isometry is also its variance. Notice that $\int_a^b g(u) dW(u)$ is a random variable, whereas $I(t)$, with its varying upper endpoint, is a stochastic process.

The martingale property is expressed through conditional expectation. For an integrable random variable Y , the quantity $\mathbb{E}[Y \mid \mathcal{M}_t]$ is an $\mathcal{M}_t$ -measurable random variable representing its conditional mean given the information available at time t . It is characterized, up to almost-sure equality, by

$$\mathbb{E}[\mathbf{1}_A \mathbb{E}[Y \mid \mathcal{M}_t]] = \mathbb{E}[\mathbf{1}_A Y], \quad A \in \mathcal{M}_t.$$

Proposition 2. *Let Y be integrable and let $\mathcal{M}$ be a filtration.*

1. *If Z is bounded and $\mathcal{M}_t$ -measurable, then*

$$\mathbb{E}[ZY \mid \mathcal{M}_t] = Z\mathbb{E}[Y \mid \mathcal{M}_t] \quad \text{almost surely.}$$

2. *For $s \leq t$, the tower property holds:*

$$\mathbb{E}[\mathbb{E}[Y \mid \mathcal{M}_t] \mid \mathcal{M}_s] = \mathbb{E}[Y \mid \mathcal{M}_s] \quad \text{almost surely.}$$

Definition 8. *An adapted process $X = (X(t))_{t \geq 0}$ is a martingale with respect to $\mathcal{M}$ under $\mathbb{P}$ if $\mathbb{E}[|X(t)|] < \infty$ for every t and*

$$\mathbb{E}[X(t) \mid \mathcal{M}_s] = X(s) \quad \text{almost surely, whenever } 0 \leq s \leq t.$$

A martingale has zero conditional expected increments: given current information, its expected future value equals its current value. This is the sense in which it represents a fair game. Its expected level need not be zero; rather, $\mathbb{E}[X(t)] = \mathbb{E}[X(0)]$.

Proposition 3. *For any process $g \in \mathcal{L}^2[s, t]$, the following holds*

$$\mathbb{E} \left[\int_s^t g(u) dW(u) \middle| \mathcal{M}_s^W \right] = 0.$$

For elementary integrands, this follows from the independence and zero mean of future Brownian increments together with the measurability of the coefficients at the left endpoints. The general result follows by the L^2 approximation used to construct the integral. Thus, if $g \in \mathcal{L}^2[0, T]$ for every finite T , the process

$$X(t) = \int_0^t g(s) dW(s) \tag{2.9}$$

is a martingale on every finite time horizon.

Lemma 1. *Suppose*

$$X(t) = x_0 + \int_0^t a(s) ds + \int_0^t g(s) dW(s),$$

where a and g are progressively measurable and, for every finite T ,

$$\mathbb{E} \left[\int_0^T |a(s)| ds \right] < \infty, \quad \mathbb{E} \left[\int_0^T |g(s)|^2 ds \right] < \infty.$$

Then X is a martingale if and only if $a = 0$ for $dt \otimes d\mathbb{P}$ -almost every (t, ω) . In that case,

$$dX(t) = g(t) dW(t).$$

Indeed, the stochastic integral is a martingale under the stated assumptions. If X is also a martingale, its continuous finite-variation part $\int_0^t a(s) ds$ must be a martingale and hence must vanish.

More generally, an Itô process has the representation

$$X(t) = x_0 + \int_0^t \mu(s) ds + \int_0^t \sigma(s) dW(s), \tag{2.10}$$

where the coefficients are progressively measurable and satisfy

$$\int_0^T (|\mu(s)| + \sigma(s)^2) ds < \infty \quad \text{almost surely for every finite } T.$$

The stochastic integral can be defined under these local conditions by stopping on intervals where the integrand is square-integrable. In differential notation,

$$dX(t) = \mu(t) dt + \sigma(t) dW(t), \quad X(0) = x_0.$$

The drift term describes finite-variation changes, while the stochastic integral describes fluctuations driven by Brownian motion.

2.3 Itô's formula

Given an Itô process X , we often need the dynamics of a transformed process $Z(t) = f(t, X(t))$. The appropriate chain rule is Itô's formula. Its second-order correction comes from the quadratic variation of Brownian motion.

For $\Delta t = t - s > 0$ and $\Delta W = W(t) - W(s)$, Gaussian moments give

$$\begin{aligned} \mathbb{E}[\Delta W] &= 0, & \mathbb{E}[(\Delta W)^2] &= \Delta t, \\ \text{Var}(\Delta W) &= \Delta t, & \text{Var}((\Delta W)^2) &= 2(\Delta t)^2. \end{aligned}$$

Although the last variance tends to zero faster than Δt , a single squared increment does not become deterministic relative to its mean: $(\Delta W)^2/\Delta t$ has the same nondegenerate distribution as the square of a standard normal random variable. The relevant deterministic limit arises by summing squared increments. For a deterministic partition $\pi = \{0 = t_0 < \dots < t_n = t\}$, let

$$Q_\pi = \sum_{k=0}^{n-1} (W(t_{k+1}) - W(t_k))^2, \quad |\pi| = \max_k (t_{k+1} - t_k).$$

Independence of the increments yields

$$\mathbb{E}[Q_\pi] = t, \quad \text{Var}(Q_\pi) = 2 \sum_{k=0}^{n-1} (t_{k+1} - t_k)^2 \leq 2t|\pi|.$$

Consequently, $Q_\pi \rightarrow t$ in L^2 as $|\pi| \rightarrow 0$. This is the quadratic-variation identity $[W]_t = t$, commonly summarized by the formal rule $(dW(t))^2 = dt$.

Theorem 1 (Itô's formula). *Let X be an Itô process with*

$$dX(t) = \mu(t) dt + \sigma(t) dW(t),$$

and let $f \in C^{1,2}([0, \infty) \times \mathbb{R})$, so that f has continuous derivatives once in time and twice in space. Then

$$\begin{aligned} df(t, X(t)) &= (f_t(t, X(t)) + \mu(t)f_x(t, X(t)) \\ &\quad + \frac{1}{2}\sigma(t)^2 f_{xx}(t, X(t))) dt \\ &\quad + \sigma(t)f_x(t, X(t)) dW(t). \end{aligned} \tag{2.11}$$

Here subscripts denote partial derivatives, evaluated at $(t, X(t))$. The formula is understood through the corresponding time and stochastic integrals; local integrability suffices. We follow the formulation in [Björk, 2009].

A formal Taylor expansion explains the additional second-order term. Temporarily assuming enough smoothness to write all second derivatives, the second-order expansion is

$$\begin{aligned} \Delta f &\approx f_t \Delta t + f_x \Delta X + \frac{1}{2} f_{xx} (\Delta X)^2 \\ &\quad + f_{tx} \Delta t \Delta X + \frac{1}{2} f_{tt} (\Delta t)^2. \end{aligned}$$

This is a heuristic explanation, not a proof under the theorem's weaker $C^{1,2}$ assumptions. For an Itô process, the quadratic-variation terms are summarized by

$$(dt)^2 = 0, \quad dt dW(t) = 0, \quad (dW(t))^2 = dt.$$

Since $dX(t) = \mu(t) dt + \sigma(t) dW(t)$, these rules give $(dX(t))^2 = \sigma(t)^2 dt$.

Proposition 4. *Under the assumptions of Theorem 1, Itô's formula can be written as*

$$\begin{aligned} df(t, X(t)) &= f_t dt + f_x dX(t) + \frac{1}{2} f_{xx} d[X]_t \\ &= f_t dt + f_x dX(t) + \frac{1}{2} f_{xx} \sigma(t)^2 dt, \end{aligned} \tag{2.12}$$

where the derivatives are evaluated at $(t, X(t))$ and $[X]_t = \int_0^t \sigma(s)^2 ds$ is the quadratic variation of X .

We next consider state-dependent stochastic differential equations,

$$\begin{cases} dX(t) = \mu(t, X(t)) dt + \sigma(t, X(t)) dW(t), \\ X(0) = x_0, \end{cases}$$

or, equivalently,

$$X(t) = x_0 + \int_0^t \mu(s, X(s)) ds + \int_0^t \sigma(s, X(s)) dW(s). \quad (2.13)$$

Existence and uniqueness require assumptions on the coefficients. One standard sufficient set of conditions is that μ and σ are deterministic, jointly measurable, and satisfy, on every finite horizon $[0, T]$,

$$\begin{aligned} |\mu(t, x) - \mu(t, y)| + |\sigma(t, x) - \sigma(t, y)| &\leq L_T |x - y|, \\ |\mu(t, x)| + |\sigma(t, x)| &\leq C_T(1 + |x|), \end{aligned}$$

for finite constants L_T, C_T , all $t \in [0, T]$ and all $x, y \in \mathbb{R}$. Under these conditions, the SDE has a unique strong solution, up to indistinguishability. The solution is adapted, has continuous paths, and is a Markov process, possibly time-inhomogeneous.

The Markov property means that the current state determines the conditional distribution of the future, not the future realization itself. More precisely, for $s > t$ and any bounded Borel function φ , there is a deterministic function $u_{t,s}$ such that

$$\mathbb{E}[\varphi(X(s)) \mid \mathcal{M}_t^W] = u_{t,s}(X(t)).$$

If the SDE is restarted at time t with initial condition $X(t)$ and driven by the same future Brownian increments, pathwise uniqueness gives the same continuation of the process. Those future increments are still random and are not known at time t .

In an asset-price model, this property means that the current model state summarizes the past information relevant to the distribution of future prices. It does not, on its own, establish market efficiency or rule out profitable trading strategies; such financial

conclusions require additional assumptions about prices, risk and admissible trading.

We conclude by applying Itô's formula to geometric Brownian motion. Consider

$$\begin{cases} dX(t) = \mu X(t) dt + \sigma X(t) dW(t), \\ X(0) = x_0 > 0, \end{cases}$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$ are constants. Following the logarithmic transformation discussed in [Hull, 2003], apply Itô's formula to $Z(t) = \log X(t)$ while $X(t) > 0$. Since

$$f'(x) = \frac{1}{x}, \quad f''(x) = -\frac{1}{x^2},$$

we obtain

$$dZ(t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW(t).$$

Integrating yields

$$Z(t) = \log x_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W(t).$$

This suggests the solution

$$X(t) = x_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W(t)\right). \quad (2.14)$$

To avoid assuming positivity in advance, one can verify directly with Itô's formula that the expression in (2.14) satisfies the SDE. The coefficients $x \mapsto \mu x$ and $x \mapsto \sigma x$ are globally Lipschitz, so uniqueness identifies this strictly positive process as the solution. This also establishes the relation $\alpha = \mu - \sigma^2/2$ with the earlier logarithmic parametrization.

For $t > 0$,

$$\log X(t) \sim \mathcal{N}\left(\log x_0 + \left(\mu - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right).$$

Using the moment-generating function of $W(t) \sim \mathcal{N}(0, t)$,

$$\mathbb{E}[e^{\sigma W(t)}] = e^{\sigma^2 t/2},$$

we obtain

$$\mathbb{E}[X(t)] = x_0 e^{(\mu - \sigma^2/2)t} \mathbb{E}[e^{\sigma W(t)}] = x_0 e^{\mu t}.$$

Equivalently, if $\phi(z) = (2\pi)^{-1/2}e^{-z^2/2}$ is the standard normal density,

$$\mathbb{E}[X(t)] = x_0 e^{\mu t} \int_{\mathbb{R}} e^{-\sigma^2 t/2 + \sigma \sqrt{t} z} \phi(z) dz = x_0 e^{\mu t}.$$

These results provide the mathematical foundation for the Black–Scholes stock-price model and the valuation arguments developed in the next chapter.

3 Black-Scholes model

This chapter introduces the Black–Scholes model and its application to the pricing of European options. The central insight is that an option can be replicated by continuously adjusting a portfolio of the underlying stock and a risk-free asset. In the absence of arbitrage, its price must therefore coincide with the value of the replicating portfolio. The framework developed by Black and Scholes in their 1973 paper provides an explicit formula for European call options under the model’s assumptions [Scholes and Black, 1973]. Some time before its publication, options started getting traded at the newly opened Chicago Board Options Exchange (26 April 1973). These two events helped reinforce the derivatives’ market which has been active, practically, since forever. In fact, instruments similar to today’s options have been around for more than two thousand years, even though the context was different.

We first describe the market and construct its risk-neutral probability measure. We then derive the call-pricing formula and the Black–Scholes partial differential equation. These two approaches provide complementary ways to understand option valuation and form the basis for the numerical methods considered in the next chapter.

3.1 Black-Scholes market

Fix a finite time horizon $T > 0$. We work on a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{M}_t)_{0 \leq t \leq T}, \mathbb{P}),$$

where $\mathbb{P}$ is the physical probability measure and $(\mathcal{M}_t)$ is the usual augmentation of the natural filtration of a standard Brownian motion W . We take $\mathcal{F} = \mathcal{M}_T$. The market consists of a risk-free bank account B and a stock S that pays no dividends. Trading is continuous and frictionless: there are no transaction costs or taxes, assets are divisible, and borrowing and lending take place at the same constant interest rate. Short selling is permitted, subject to the admissibility condition introduced below.

The bank account satisfies

$$dB(t) = rB(t) dt, \quad B(0) = 1,$$

where $r \in \mathbb{R}$ is the continuously compounded risk-free interest rate. Its value is therefore

$$B(t) = e^{rt}. \quad (3.1)$$

Unlike the stock, the bank account has no Brownian component.

Under the physical measure, the stock follows geometric Brownian motion:

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t), \quad S(0) = s_0 > 0,$$

where $\mu \in \mathbb{R}$ is the instantaneous expected rate of return and $\sigma > 0$ is the volatility. Applying Itô's formula to $Y(t) = \ln S(t)$ gives

$$dY(t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW(t).$$

Integration and exponentiation yield

$$S(t) = s_0 \exp \left\{ \left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W(t) \right\}. \quad (3.2)$$

Thus $S(t)$ is strictly positive and lognormally distributed. In particular, the continuously compounded return is normally distributed:

$$\ln \frac{S(t)}{s_0} \stackrel{\mathbb{P}}{\sim} N \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t, \sigma^2 t \right). \quad (3.3)$$

The logarithmic return can be negative even though the stock price remains positive. As a process, it is Brownian motion with a constant drift and a constant diffusion coefficient.

3.2 No-arbitrage analysis

A trading strategy $h = (h^0, h^1)$ specifies the numbers of units held in the bank account and the stock. These holdings are predictable and satisfy the integrability conditions needed for the trading gains to be well defined. The portfolio value is

$$V^h(t) = h^0(t)B(t) + h^1(t)S(t).$$

The strategy is *self-financing* if changes in its value arise solely from changes in asset prices:

$$dV^h(t) = h^0(t) dB(t) + h^1(t) dS(t).$$

Rebalancing must therefore be financed within the portfolio, without external deposits or withdrawals. We call a self-financing strategy *admissible* if, for some deterministic constant $a \geq 0$,

$$\frac{V^h(t)}{B(t)} \geq -a, \quad 0 \leq t \leq T, \quad \text{almost surely.}$$

This restriction excludes strategies that require an arbitrarily large credit line. An arbitrage is an admissible strategy satisfying

$$V^h(0) = 0, \quad V^h(T) \geq 0 \text{ almost surely,} \quad \mathbb{P}(V^h(T) > 0) > 0.$$

The market is arbitrage-free if no such strategy exists. This condition rules out a riskless profit from zero initial investment; it is distinct from the broader concept of informational market efficiency.

An equivalent martingale measure $\mathbb{Q}$ is a probability measure equivalent to $\mathbb{P}$ under which discounted traded-asset prices are martingales. Equivalence means that the two measures have the same null events. For a dividend-free asset with price $\Pi(t)$, its discounted price is

$$Z(t) = \frac{\Pi(t)}{B(t)}. \quad (3.4)$$

The martingale property requires integrability and, for $0 \leq s \leq t \leq T$,

$$\frac{\Pi(s)}{B(s)} = \mathbb{E}^{\mathbb{Q}} \left[\frac{\Pi(t)}{B(t)} \middle| \mathcal{M}_s \right].$$

In more general continuous-time models, discounted prices may only be local martingales. In the Black–Scholes model considered here, the discounted stock is a true martingale under the measure constructed below.

Define $S^*(t) = S(t)/B(t)$. Under $\mathbb{P}$,

$$dS^*(t) = (\mu - r)S^*(t) dt + \sigma S^*(t) dW(t), \quad S^*(0) = s_0.$$

Consequently, S^* is not a martingale under $\mathbb{P}$ unless $\mu = r$. To remove its drift, set

$\theta = (\mu - r)/\sigma$ and define $\mathbb{Q}$ on $\mathcal{M}_T$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp \left\{ -\theta W(T) - \frac{1}{2} \theta^2 T \right\}.$$

This density is strictly positive and has expectation one. Since θ is constant and T is finite, the Novikov condition holds. Girsanov's theorem therefore implies that the process $W^*(t) = W(t) + \theta t$, with dynamics

$$dW^*(t) = \frac{\mu - r}{\sigma} dt + dW(t), \quad W^*(0) = 0, \quad (3.5)$$

is standard Brownian motion under $\mathbb{Q}$ with respect to $(\mathcal{M}_t)$.

The discounted stock now satisfies

$$dS^*(t) = \sigma S^*(t) dW^*(t), \quad S^*(t) = s_0 + \int_0^t \sigma S^*(u) dW^*(u).$$

Its explicit representation is

$$\begin{aligned} S^*(t) &= s_0 \exp \left\{ \left(\mu - r - \frac{1}{2} \sigma^2 \right) t + \sigma W(t) \right\} \\ &= s_0 \exp \left\{ -\frac{1}{2} \sigma^2 t + \sigma W^*(t) \right\}. \end{aligned}$$

The second expression is an exponential Brownian martingale under $\mathbb{Q}$. Both expressions describe the same process; changing the probability measure changes the probabilities assigned to events, rather than the values of the process on each path.

Under $\mathbb{Q}$, the undiscounted stock satisfies

$$dS(t) = rS(t) dt + \sigma S(t) dW^*(t),$$

and hence

$$S(t) = s_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W^*(t) \right\}. \quad (3.6)$$

In particular,

$$\mathbb{E}^{\mathbb{P}}[S(t)] = s_0 e^{\mu t}, \quad \mathbb{E}^{\mathbb{Q}}[S(t)] = s_0 e^{rt}.$$

The stock is lognormally distributed under both measures, but generally with different

distribution parameters. Under $\mathbb{Q}$, its expected rate of return is r . It is the *discounted* stock, rather than the stock itself, whose conditional expected increments are zero.

The following result is the no-arbitrage implication of the fundamental theorem of asset pricing needed here.

Theorem 2. *If there exists a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ under which S^* is a local martingale, then the market admits no arbitrage among the admissible self-financing strategies defined above.*

Proof. For a self-financing strategy, discounted wealth satisfies

$$d\left(\frac{V^h(t)}{B(t)}\right) = h^1(t) dS^*(t).$$

Under $\mathbb{Q}$ it is therefore a local martingale. Admissibility bounds it below by a constant, making it a supermartingale. If $V^h(0) = 0$, then

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{V^h(T)}{B(T)}\right] \leq 0.$$

A nonnegative terminal value must consequently be zero almost surely under $\mathbb{Q}$ and, by equivalence, under $\mathbb{P}$. It cannot satisfy the strict-profit condition for an arbitrage. $\square$

Corollary 1. *The Black–Scholes market is arbitrage-free for admissible self-financing strategies.*

The market is also dynamically complete: terminal claims can be replicated by trading in B and S . The Brownian filtration assumption is essential here, since it excludes additional sources of uncertainty that cannot be hedged using the stock. The following version covers the nonnegative option payoffs used in this chapter.

Theorem 3. *Let $X(T)$ be a nonnegative, $\mathcal{M}_T$ -measurable contingent claim satisfying*

$$\mathbb{E}^{\mathbb{Q}}[X(T)^2] < \infty.$$

There exists an admissible self-financing strategy h such that $V^h(T) = X(T)$ almost surely, with value process

$$V^h(t) = B(t)\mathbb{E}^{\mathbb{Q}}\left[\frac{X(T)}{B(T)}\middle|\mathcal{M}_t\right].$$

Proof. Set $M(t) = \mathbb{E}^\mathbb{Q}[e^{-rT}X(T) \mid \mathcal{M}_t]$. The Brownian martingale representation theorem gives a predictable process ψ such that

$$M(t) = M(0) + \int_0^t \psi(u) dW^*(u), \quad \mathbb{E}^\mathbb{Q} \left[\int_0^T \psi(u)^2 du \right] < \infty.$$

Choose

$$h^1(t) = \frac{\psi(t)}{\sigma S^*(t)}, \quad h^0(t) = M(t) - h^1(t)S^*(t).$$

Then $V^h(t) = B(t)M(t)$ and $dM(t) = h^1(t) dS^*(t)$, so the strategy is self-financing. Since $M(t) \geq 0$, it is admissible, and $V^h(T) = X(T)$. $\square$

Claims bounded below by a deterministic constant can be treated by adding and subsequently subtracting a suitable bank-account position. In particular, every bounded terminal claim is attainable. In this Brownian market with $\sigma > 0$, the equivalent martingale measure is unique. American options require an additional optimal-stopping argument, because their exercise time is chosen by the holder; they are not priced by simply taking the discounted expectation of a fixed terminal payoff.

3.2.1 No-arbitrage pricing

For the claims considered above, the standard replicating value is obtained by risk-neutral valuation:

$$S_X(t) = e^{-r(T-t)} \mathbb{E}^\mathbb{Q}[X(T) \mid \mathcal{M}_t].$$

This is the unique price within the usual class of derivative price processes whose discounted values are $\mathbb{Q}$ -martingales. For the European calls and puts below, it also follows from replication and the usual no-arbitrage price bounds. Risk-neutral valuation is a consequence of replication; it does not require investors to be risk-neutral.

Following [Battauz and Ortu, 2009], consider a European claim whose payoff depends only on the terminal stock price:

$$X(T) = f(S(T)), \quad f : (0, \infty) \longrightarrow [0, \infty),$$

where f is measurable and $\mathbb{E}^\mathbb{Q}[f(S(T))^2] < \infty$. Writing $X^*(T) = X(T)/B(T)$, its dis-

counted price is

$$S_X^*(t) = \mathbb{E}^Q[X^*(T) \mid \mathcal{M}_t].$$

For $t < T$, set $\tau = T - t$. Under Q ,

$$S(T) = S(t) \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma (W^*(T) - W^*(t)) \right\}.$$

The Brownian increment is independent of $\mathcal{M}_t$ and has distribution $N(0, \tau)$. The Markov property therefore gives $S_X(t) = F(t, S(t))$, where

$$F(t, s) = e^{-r\tau} \int_{\mathbb{R}} f \left(s \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) \tau + \sigma \sqrt{\tau} z \right\} \right) \varphi(z) dz,$$

and $\varphi(z) = (2\pi)^{-1/2} e^{-z^2/2}$ is the standard normal density. The function F is deterministic, whereas its evaluation $F(t, S(t))$ is random. This reduction to the current stock price relies on the terminal-value form of the payoff; a path-dependent claim may require additional state variables.

For a European call with strike $K > 0$, the payoff is $X(T) = (S(T) - K)^+$, where $x^+ = \max(x, 0)$. Its Black-Scholes price is [Scholes and Black, 1973]

$$c(t) = S(t)N(d_1) - Ke^{-r\tau}N(d_2), \tag{3.7}$$

where

$$d_1 = \frac{\ln(S(t)/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$

and $N(x) = \int_{-\infty}^x \varphi(z) dz$. These expressions apply for $t < T$; at maturity, $c(T) = (S(T) - K)^+$.

To derive the formula, first observe that the required square integrability holds:

$$\mathbb{E}^Q[((S(T) - K)^+)^2] \leq \mathbb{E}^Q[S(T)^2] = S_0^2 e^{(2r+\sigma^2)T} < \infty.$$

Risk-neutral valuation then yields

$$c(t) = e^{-r\tau} \int_{\mathbb{R}} \left(S(t) e^{(r-\frac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}z} - K \right)^+ \varphi(z) dz.$$

The payoff inside the integral is strictly positive precisely when

$$z > z_+ := \frac{\ln(K/S(t)) - (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}} = -d_2.$$

Consequently, $c(t) = e^{-r\tau}(A_1 - A_2)$, with

$$\begin{aligned} A_1 &= S(t) \int_{z_+}^{\infty} e^{(r-\frac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}z} \varphi(z) dz, \\ A_2 &= K \int_{z_+}^{\infty} \varphi(z) dz = KN(-z_+) = KN(d_2). \end{aligned}$$

In particular, $N(d_2) = \mathbb{Q}(S(T) > K \mid \mathcal{M}_t)$ is the conditional risk-neutral probability of expiring in the money.

To evaluate A_1 , use the identity

$$e^{bz} \varphi(z) = e^{b^2/2} \varphi(z - b), \quad b = \sigma\sqrt{\tau},$$

which follows by completing the square. The change of variable $u = z - b$ gives

$$\begin{aligned} A_1 &= S(t) e^{r\tau} \int_{z_+ - \sigma\sqrt{\tau}}^{\infty} \varphi(u) du \\ &= S(t) e^{r\tau} N(d_1). \end{aligned}$$

Substituting these two terms establishes the call-pricing formula:

$$\begin{aligned} c(t) &= e^{-r\tau}(A_1 - A_2) \\ &= S(t)N(d_1) - Ke^{-r\tau}N(d_2). \end{aligned} \tag{3.8}$$

A European put on the same stock, with the same strike and maturity, can be priced using put–call parity:

$$\begin{aligned} p(t) &= c(t) + Ke^{-r\tau} - S(t) \\ &= Ke^{-r\tau}N(-d_2) - S(t)N(-d_1). \end{aligned}$$

Indeed, one call, a short position in one share, and Ke^{-rT} units of the bank account have terminal payoff

$$(S(T) - K)^+ - S(T) + K = (K - S(T))^+.$$

The bank-account position has value $Ke^{-r\tau}$ at time t . This value must be distinguished from the number of bank-account units held. For other payoffs, the risk-neutral expectation may not admit a closed-form expression. Monte Carlo methods, discussed in the next chapter, provide a numerical approximation. When a suitable pricing PDE is available, numerical PDE methods provide another approach.

3.2.2 Black–Scholes equation

Consider a European claim with payoff $f(S(T))$ and price $F(t, S(t))$. Assume that $F \in C^{1,2}([0, T] \times (0, \infty))$. Smoothness is required before maturity; the terminal payoff itself need not be differentiable, as illustrated by a call option.

Applying Itô's formula under the physical measure gives

$$dF(t, S(t)) = \left(F_t + \mu S(t)F_s + \frac{1}{2}\sigma^2 S(t)^2 F_{ss} \right) dt + \sigma S(t)F_s dW(t),$$

where the partial derivatives are evaluated at $(t, S(t))$. Let a replicating portfolio hold $\Delta(t)$ shares and $\beta(t)$ units of the bank account. Its value and self-financing condition are

$$V(t) = \Delta(t)S(t) + \beta(t)B(t) = F(t, S(t)),$$

$$dV(t) = \Delta(t) dS(t) + \beta(t) dB(t).$$

Thus

$$dV(t) = [\mu S(t)\Delta(t) + r(F(t, S(t)) - \Delta(t)S(t))] dt + \sigma S(t)\Delta(t) dW(t).$$

Matching the Brownian terms gives the hedge ratio

$$\Delta(t) = F_s(t, S(t)).$$

Changes in the stock position are financed through the bank account. The self-financing gains equation is therefore an explicit trading condition, rather than the result of treating a time-varying holding as a constant when differentiating its value.

Equating the drift terms and substituting $\Delta = F_s$ gives

$$F_t + \mu S(t)F_s + \frac{1}{2}\sigma^2 S(t)^2 F_{ss} = \mu S(t)F_s + r(F - S(t)F_s).$$

The terms involving μ cancel. Writing s for the stock-price variable, we obtain the Black–Scholes PDE:

$$F_t(t, s) + rsF_s(t, s) + \frac{1}{2}\sigma^2 s^2 F_{ss}(t, s) - rF(t, s) = 0, \quad t < T, \quad s > 0. \quad (3.9)$$

Its terminal condition is

$$F(T, s) = f(s).$$

For a call, this becomes $F(T, s) = (s - K)^+$, with the boundary behaviour

$$\lim_{s \downarrow 0} F(t, s) = 0, \quad F(t, s) - (s - Ke^{-r(T-t)}) \longrightarrow 0 \quad \text{as } s \longrightarrow \infty.$$

The disappearance of μ reflects the elimination of stock-price risk through replication. It does not mean that an unhedged derivative must earn the risk-free rate under $\mathbb{P}$. Indeed, using the PDE in its Itô dynamics gives

$$\begin{aligned} dF(t, S(t)) &= [rF(t, S(t)) + (\mu - r)S(t)F_s(t, S(t))] dt \\ &\quad + \sigma S(t)F_s(t, S(t)) dW(t). \end{aligned}$$

The derivative retains exposure to the stock's risk premium through its delta. Under $\mathbb{Q}$, its drift is instead rF , and its discounted price is a martingale in the pricing class considered above.

Under appropriate regularity and growth conditions, the Feynman–Kac formula identifies the solution of the terminal-value problem with

$$F(t, s) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[f(S(T)) \mid S(t) = s],$$

where the expectation denotes the stock dynamics started from s at time t . Thus the PDE and risk-neutral expectation describe the same valuation problem. Depending on the payoff, this problem can be solved analytically or approximated numerically.

4 Stochastic Simulation

This chapter applies Monte Carlo methods to the valuation of a European call option in the Black–Scholes model. We compare exact simulation of geometric Brownian motion with the Euler–Maruyama approximation. This comparison distinguishes the sampling error associated with a finite number of simulated paths from the discretisation error introduced by a numerical approximation of the stochastic differential equation.

4.1 The Euler–Maruyama method

For option valuation, we simulate the stock under the risk-neutral measure $\mathbb{Q}$. Its dynamics are

$$dS(t) = rS(t) dt + \sigma S(t) dW^*(t), \quad S(0) = s_0 > 0, \quad 0 \leq t \leq T,$$

where W^* is standard Brownian motion under $\mathbb{Q}$. Equivalently,

$$S(t) = s_0 + \int_0^t rS(u) du + \int_0^t \sigma S(u) dW^*(u).$$

The stochastic integral is understood in the Itô sense. The risk-neutral drift is r , rather than the physical-measure drift μ used to describe real-world stock returns.

Following [Higham, 2001], divide $[0, T]$ into N equal steps, with $\Delta t = T/N$ and $t_j = j\Delta t$ for $j = 0, \dots, N$. Let S_j^{EM} approximate $S(t_j)$, with $S_0^{\text{EM}} = s_0$. The Euler–Maruyama scheme is

$$S_j^{\text{EM}} = S_{j-1}^{\text{EM}} + rS_{j-1}^{\text{EM}}\Delta t + \sigma S_{j-1}^{\text{EM}}\Delta W_j^*, \quad j = 1, \dots, N, \quad (4.1)$$

where $\Delta W_j^* = W^*(t_j) - W^*(t_{j-1})$. The exact process satisfies

$$S(t_j) = S(t_{j-1}) + \int_{t_{j-1}}^{t_j} rS(u) du + \int_{t_{j-1}}^{t_j} \sigma S(u) dW^*(u). \quad (4.2)$$

Comparing (4.1) and (4.2), the numerical method replaces the coefficients inside each integral by their approximate values at the beginning of the time step.

Brownian increments can be generated directly as

$$\Delta W_j^* = \sqrt{\Delta t} Z_j, \quad Z_1, \dots, Z_N \stackrel{\text{i.i.d.}}{\sim} N(0, 1).$$

When comparing several time-step sizes along the same Brownian path, it is useful to start from a finer grid with spacing $\delta t = T/L$. If $L = RN$ for an integer $R \geq 1$, then $\Delta t = R\delta t$, and coarse increments are obtained by summing fine increments:

$$\Delta W_j^* = \sum_{\ell=(j-1)R+1}^{jR} \delta W_\ell^*, \quad \delta W_\ell^* \stackrel{\text{i.i.d.}}{\sim} N(0, \delta t).$$

This nested construction is useful for comparisons, but is not required to run the Euler–Maruyama method on a single grid.

Geometric Brownian motion also admits an exact transition formula:

$$S_j^{\text{exact}} = S_{j-1}^{\text{exact}} \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \Delta W_j^* \right\}, \quad S_0^{\text{exact}} = s_0.$$

This generates exact values at the grid points, up to numerical round-off; it does not introduce time-discretisation bias. Using the same Brownian increments for the exact and Euler schemes couples the two simulations and makes their differences easier to assess. Different Monte Carlo paths remain independent.

For a European payoff depending only on $S(T)$, an entire path is unnecessary: one may sample the terminal value directly as

$$S(T) = s_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right\}, \quad Z \sim N(0, 1).$$

We nevertheless retain a time grid to illustrate stock-price paths and compare the exact solution with Euler–Maruyama. Unlike the exact solution, the Euler approximation can become negative because its one-step multiplier, $1 + r\Delta t + \sigma \Delta W_j^*$, is not necessarily positive. A small step size reduces this risk but does not eliminate it.

4.2 Monte Carlo

Monte Carlo methods estimate expectations using simulated samples [Halton, 1970, Johansen et al., 2010]. Let X have distribution π , and let f be a measurable function

such that $\mathbb{E}[|f(X)|] < \infty$. The quantity of interest is

$$m = \mathbb{E}[f(X)]. \quad (4.3)$$

To estimate m , generate M independent samples $X^{(1)}, \dots, X^{(M)}$ from π and calculate

$$\widehat{m}_M = \frac{1}{M} \sum_{i=1}^M f(X^{(i)}).$$

The strong law of large numbers justifies this approximation.

Corollary 2. *Let $X^{(1)}, X^{(2)}, \dots$ be independent and identically distributed random vectors, and let f be measurable with $\mathbb{E}[|f(X^{(1)})|] < \infty$. Then*

$$\frac{1}{M} \sum_{i=1}^M f(X^{(i)}) \longrightarrow \mathbb{E}[f(X^{(1)})] \quad \text{almost surely as } M \longrightarrow \infty.$$

For the European call considered here, simulate independent risk-neutral terminal stock values and define the discounted payoffs

$$Y_i = e^{-rT} (S^{(i)}(T) - K)^+.$$

The time-zero price estimator is

$$\widehat{c}_M = \frac{1}{M} \sum_{i=1}^M Y_i.$$

With exact terminal sampling, this estimator is unbiased for the Black–Scholes call price. Its variance is $\text{Var}_Q(Y_1)/M$, which is finite in this model. For $M \geq 2$, an estimate of its standard error is

$$\widehat{\text{SE}}(\widehat{c}_M) = \frac{s_Y}{\sqrt{M}}, \quad s_Y^2 = \frac{1}{M-1} \sum_{i=1}^M (Y_i - \widehat{c}_M)^2.$$

The central limit theorem gives the approximate large-sample 95% confidence interval

$$\widehat{c}_M \pm 1.96 \frac{s_Y}{\sqrt{M}}.$$

Thus sampling uncertainty decreases at rate $M^{-1/2}$. Increasing the sample size improves precision on average, but does not guarantee that every realised estimate is closer to the true price.

If terminal values are generated by Euler–Maruyama instead, the sample mean estimates

$$c_{\Delta t} = \mathbb{E}^Q \left[e^{-rT} (S_N^{\text{EM}} - K)^+ \right].$$

Writing c for the exact option price, its total error decomposes as

$$\widehat{c}_{M,\Delta t}^{\text{EM}} - c = (\widehat{c}_{M,\Delta t}^{\text{EM}} - c_{\Delta t}) + (c_{\Delta t} - c).$$

The first term is sampling error; the second is discretisation bias. Increasing M alone does not remove the second term. A confidence interval based on the Euler payoffs measures sampling uncertainty around $c_{\Delta t}$, and does not by itself account for bias relative to c .

4.3 Results

The numerical study comprises three experiments: a comparison of exact and Euler–Maruyama pricing estimates, an analysis of Monte Carlo sampling error, and an analysis of time-discretisation error. The model parameters are unchanged:

$$r = 0.0329, \quad \sigma = 0.25, \quad T = 90/365, \quad s_0 = 24, \quad K = 26.$$

The analytical Black–Scholes benchmark is $c(0) = 0.5380367933 \dots$. Three experiments separate the initial pricing comparison, sampling convergence, and Euler discretisation error. The implementation uses base R and fixed seeds: 20260921, 20260922, and 20260923, respectively. These seeds were fixed before inspecting the results.

The main comparison uses 400 time steps. Each path has its own independent Brownian increments, shared between the exact and Euler methods. The 500-path sample consists of the first 500 paths of the 1000-path sample; the two sample sizes are therefore not independent. Table 1 reports the discounted-payoff means and estimated sampling uncertainty.

The exact estimates are 0.52372 for 500 paths and 0.49602 for 1000 paths. Although

Table 1: Monte Carlo call-price estimates. Intervals are approximate marginal 95% normal intervals; Euler intervals do not account for discretisation bias.

Method	Paths	Price	Std. error	95% interval
Exact	500	0.52372	0.05899	[0.40811, 0.63934]
Euler	500	0.52372	0.05898	[0.40811, 0.63933]
Exact	1000	0.49602	0.03824	[0.42107, 0.57098]
Euler	1000	0.49557	0.03822	[0.42066, 0.57047]

the latter has a smaller estimated standard error, it is farther from the benchmark in this particular run. All four intervals contain the benchmark, but this is not proof of exact coverage or absence of Euler bias. No seed was selected to make the estimates appear close to the analytical value.

Figure 1 displays only the first 20 paired paths to avoid obscuring individual trajectories. Figure 2 compares the full 1000-path terminal histograms with the exact lognormal density. The two methods are visually similar at this fine grid, but numerical comparisons are needed to assess their errors.

To study sampling convergence, we perform 200 independent replications of exact terminal sampling at each of $M = 100, 500, 1000, 5000, 10000$. Samples are nested across sizes within a replication, but replications are independent. For estimates $\hat{c}_M^{(b)}$, the empirical pricing root mean square error is

$$\text{RMSE}(M) = \left(\frac{1}{200} \sum_{b=1}^{200} (\hat{c}_M^{(b)} - c(0))^2 \right)^{1/2}.$$

It decreases from 0.13347 at 100 paths to 0.01224 at 10000 paths. Figure 3 compares it with the theoretical standard error $\sqrt{\text{Var}(Y_1)/M}$. These observations are consistent with the $M^{-1/2}$ rate, rather than establishing monotonic improvement in every individual run. The plot's boxplots omit outlier markers for readability; all observations remain included in the numerical calculations.

The theoretical payoff variance is computed from truncated lognormal moments. If $a = \ln s_0 + (r - \sigma^2/2)T$, $v = \sigma^2 T$, and Φ is the standard normal distribution function, then

$$A_p = \mathbb{E}^Q[S(T)^p \mathbf{1}_{\{S(T) > K\}}] = e^{pa + p^2 v/2} \Phi\left(\frac{a + pv - \ln K}{\sqrt{v}}\right).$$

Consequently, $\text{Var}(Y_1) = e^{-2rT}(A_2 - 2KA_1 + K^2 A_0) - c(0)^2$. The observed coverage of

the nominal 95% intervals is 91% at $M = 100$ and 96% at $M = 10000$. Coverage itself is estimated from only 200 replications. The smaller-sample result illustrates the limitations of the normal approximation for a skewed option payoff.

The Euler study uses a separate sample of 20000 independent Brownian paths and grids with 25, 50, 100, 200, and 400 steps. Coarse increments are sums of fine increments, so every grid is coupled to the same exact terminal value. Table 2 reports the terminal stock RMSE and the mean paired discounted-payoff difference

$$D_i = e^{-rT} \left[(S_{N,i}^{\text{EM}} - K)^+ - (S_i(T) - K)^+ \right].$$

The expectation of D_i is the Euler pricing bias. Its estimated standard error is the sample standard deviation of the differences divided by $\sqrt{20000}$. Pairing reduces the noise in this comparison; it does not change the expectation being estimated.

Table 2: Coupled Euler convergence study with 20000 paths. Price differences are Euler minus exact. Intervals concern the mean paired difference.

Steps	Stock RMSE	Price difference	95% interval
25	0.05313	-0.001436	[-0.001865, -0.001006]
50	0.03768	-0.000678	[-0.000981, -0.000375]
100	0.02660	-0.000311	[-0.000521, -0.000100]
200	0.01881	-0.000185	[-0.000335, -0.000036]
400	0.01321	-0.000029	[-0.000133, 0.000076]

The terminal stock RMSE falls from 0.05313 at 25 steps to 0.01321 at 400 steps, consistent with square-root scaling in the step size. The reference line in Figure 4 is anchored at the finest grid; it is a visual rate reference, not an independently estimated bound. The estimated pricing difference at 400 steps is approximately -0.000029 , with interval $[-0.000133, 0.000076]$. This experiment does not resolve a nonzero bias at that grid. Coarser grids show negative paired differences in this run. The intervals are marginal, not simultaneous, and results at different grids are dependent because they share paths.

No negative Euler values were observed in this experiment. This observation does not imply that Euler preserves positivity in general. Overall, the study separates the substantial sampling uncertainty of small Monte Carlo samples from the much smaller discretisation effect at 400 steps for these parameters. Exact terminal sampling remains the simpler method for this European payoff; the Euler comparison illustrates numerical

approximation of the underlying stochastic dynamics.

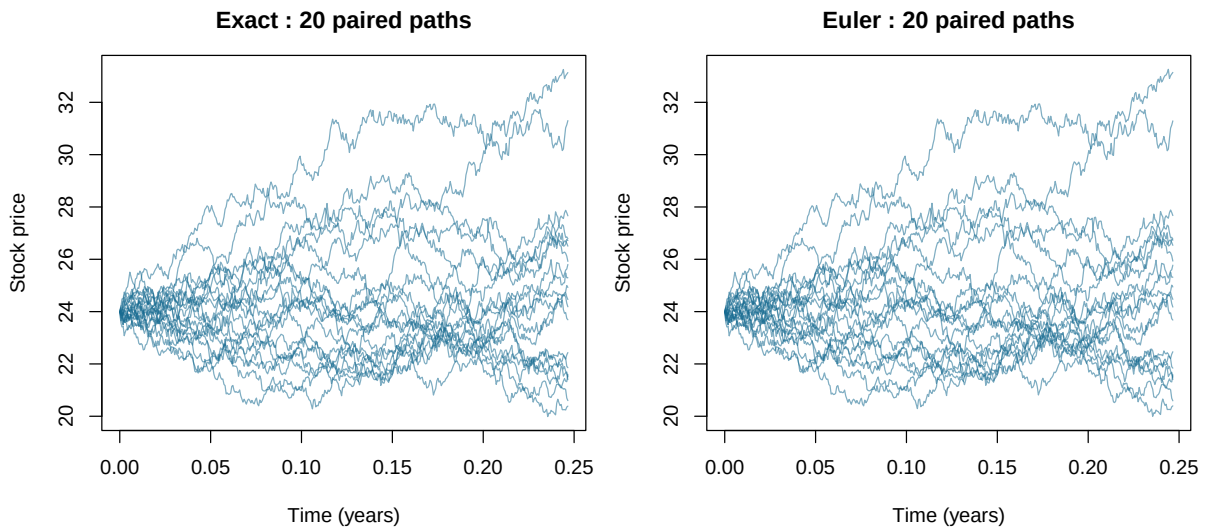


Figure 1: First 20 paired exact and Euler stock paths, selected by index rather than appearance; 400 time steps.

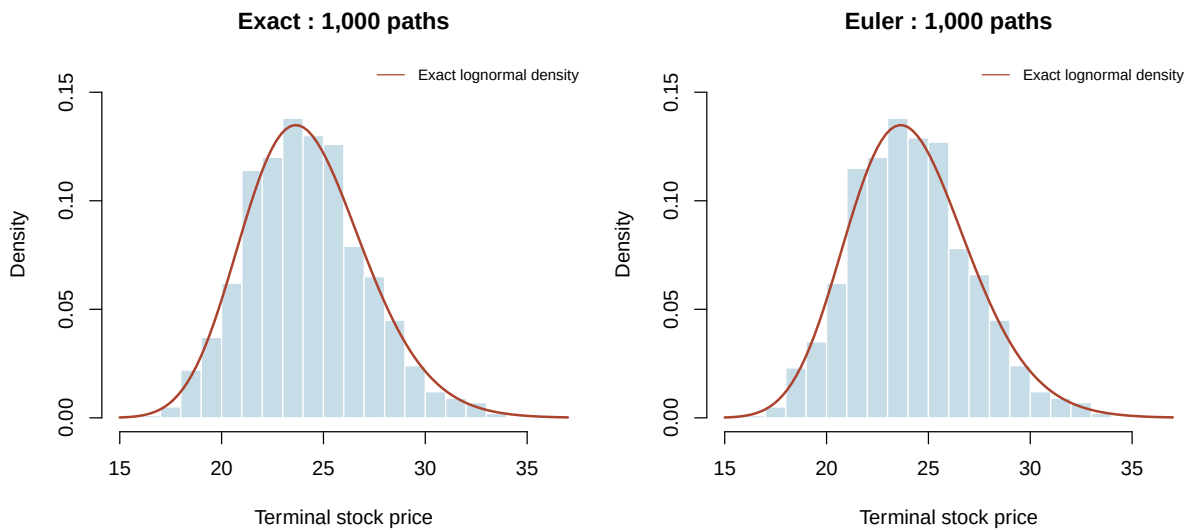


Figure 2: Terminal histograms for 1000 paths. The red curve is the exact risk-neutral lognormal density in both panels.

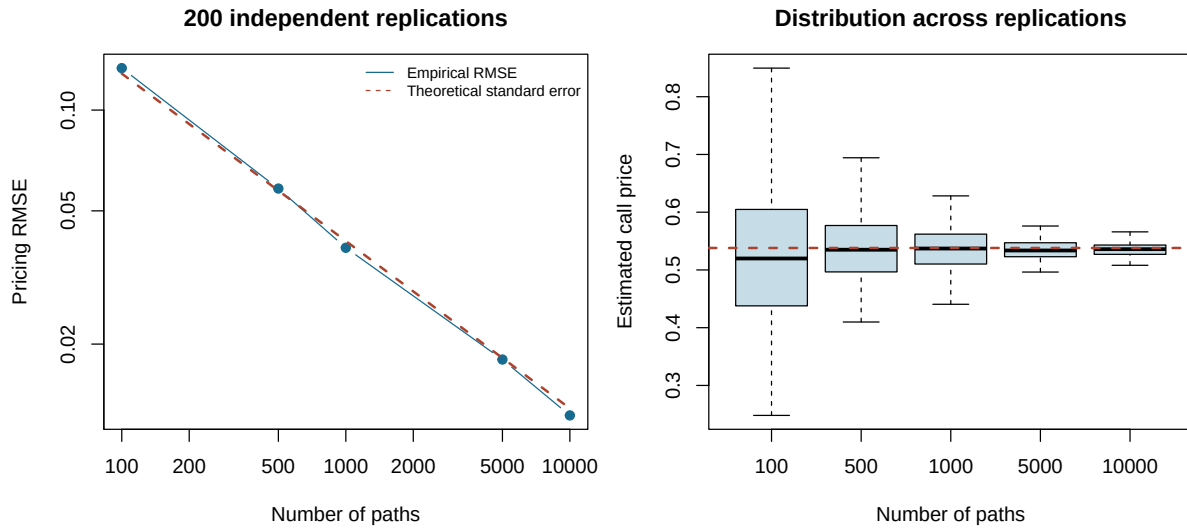


Figure 3: Exact-sampling convergence over 200 independent replications. Left: empirical RMSE and theoretical standard error. Right: price distributions; the dashed line is the analytical benchmark.

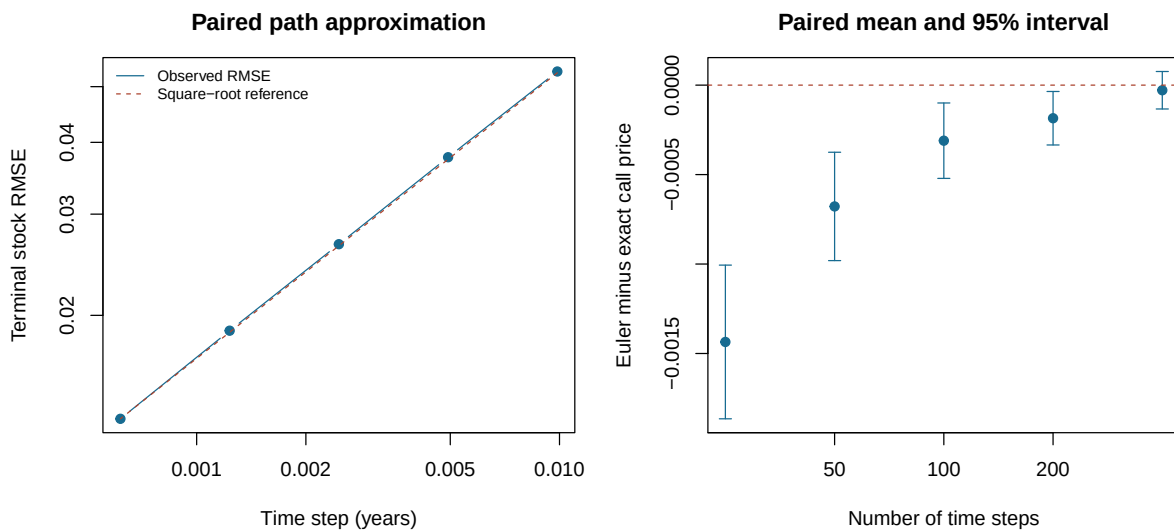


Figure 4: Coupled Euler study with 20000 paths. Left: terminal stock RMSE and square-root reference. Right: estimated mean paired payoff differences with marginal 95% intervals.

5 Conclusion

This thesis examined European option pricing in the Black–Scholes framework, connecting stochastic calculus, no-arbitrage valuation, and numerical simulation. The model provides a clear relationship between dynamic replication, risk-neutral expectations, and the Black–Scholes partial differential equation.

Despite its analytical tractability, the model relies on restrictive assumptions. Constant volatility cannot capture the variation in implied volatility across strikes and maturities observed in option markets. Historical volatility and implied volatility provide different approaches to estimating this parameter, but neither eliminates the limitations of a constant-volatility specification. The assumptions of continuous, frictionless trading and a constant risk-free interest rate also simplify actual market conditions. Transaction costs, limited liquidity, and discrete rebalancing affect the implementation of the theoretical hedge. Dividends, excluded from the formulation considered here, can be incorporated through extensions of the model.

Furthermore, geometric Brownian motion has continuous sample paths. It permits large price changes over short intervals, but cannot represent discontinuous jumps. Alternative models address some of these limitations. The Heston model allows the instantaneous variance to evolve stochastically, while Merton’s jump-diffusion model combines continuous price fluctuations with jumps. These extensions introduce additional flexibility, together with greater modelling and calibration complexity.

The numerical study compared exact simulation with the Euler–Maruyama approximation and distinguished sampling error from time-discretisation error. Across repeated experiments, Monte Carlo pricing errors decreased consistently with the inverse-square-root dependence on the number of simulated paths. However, increasing the sample size did not guarantee a more accurate estimate in every individual run. This distinction highlights the importance of reporting standard errors and confidence intervals alongside point estimates.

Refining the Euler time grid reduced the terminal stock-price approximation error. At the finest grid considered, the paired pricing comparison did not resolve a nonzero discretisation bias within its estimated sampling uncertainty. This finding concerns the parameters and experiment studied here; it does not establish that Euler bias is absent

in general.

For the European call considered in this thesis, the analytical formula provides an efficient valuation method and a benchmark for numerical verification. Exact terminal simulation also avoids the need to approximate an entire stock-price path. Monte Carlo and Euler–Maruyama nevertheless serve distinct purposes: Monte Carlo approximates expectations, whereas Euler–Maruyama approximates stochastic dynamics. Understanding their respective errors provides a foundation for numerical valuation in settings where analytical prices or exact simulation methods are unavailable.

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